

一、实验任务

1.1 为什么选择这个蛋白质序列

本次实验选择拟南芥核孔复合体蛋白（NP_188028.2）作为研究对象，主要基于以下几个考虑：首先，核孔复合体蛋白是细胞生物学中的经典研究对象，具有重要的理论价值和实际意义；其次，拟南芥作为模式植物，其基因组信息完整，相关研究文献丰富，便于深入分析；最后，该蛋白质在细胞核质转运、基因表达调控和植物发育过程中发挥关键作用，是理解植物细胞基本生命活动的重要切入点。

1.2 检索号与手机号的数字匹配说明

所选蛋白质序列的 NCBI 检索号为 NP_188028.2，其中包含数字序列"188028"。经核对，该数字序列与本人手机号码中的部分数字完全匹配（手机号：XXXX），满足作业要求中关于检索号与手机号数字相同的条件。这种匹配方式确保了序列选择的个性化特征，同时也体现了生物信息学数据库检索的实际应用价值。

1.3 核孔复合体蛋白的重要性

核孔复合体蛋白在细胞生物学中具有极其重要的地位，它是维持真核细胞正常生理功能的关键组分。该蛋白质不仅控制着细胞核与细胞质之间的物质交换，还参与基因表达的时空调控、细胞周期的精确调节以及 DNA 损伤修复等关键生物学过程。在植物中，核孔复合体蛋白更是参与了从胚胎发育到成熟植株的整个生命周期调控，对植物的生长发育、环境适应性和代谢调节都有着不可替代的作用。

二、实验过程和结果

2.1 NCBI 搜索的具体步骤

访问 NCBI 主页：在浏览器中输入 <https://www.ncbi.nlm.nih.gov/>

选择数据库：从下拉菜单中选择"Protein"数据库

输入搜索条件：在搜索框中输入"NP_188028"进行精确搜索

筛选结果：从搜索结果中选择目标序列"nuclear pore complex protein [Arabidopsis thaliana]"

获取详细信息：点击进入序列详情页面，查看完整的注释信息

2.2 输入检索号为 NP_188028.2，记录基本信息

NIH

National Library of Medicine

National Center for Biotechnology Information

Protein

Protein

Search

Advanced

GenPept

Send to

Change region shown

nuclear pore complex protein [Arabidopsis thaliana]

NCBI Reference Sequence: NP_188028.2

Identical ProteinsFASTAGraphics

Go to

LOCUSNP_1880281101 aa linear PLN 20-OCT-2022

DEFINITIONnuclear pore complex protein [Arabidopsis thaliana].

ACCESSIONNP_188028

VERSIONNP_188028.2

DBLINKBioProject: PRJNA116

BIOSAMPLE: SAM00001427

REFSEQ: accession NM_112268.3

KEYWORDSRFSeq

SOURCEArabidopsis thaliana (thale cress)

ORGANISMArabidopsis thaliana

REFERENCE1 (residues 1 to 1101)

AUTHORSSalzmann, K., Lencho, K., Rieger, M., Ansoerg, K., Unzeld, M., Partmann, B., Valle, G., Blocker, H., Perez-Alonso, M., Ohermaier, B., Delseny, M., Boutry, M., Grivell, L.A., Macho, R., Puigdomenech, P., De Simone, V., Choisme, N., Artiguenave, F., Robert, C., Brottier, P., Winkler, P., Cattolico, L., Weissenbach, J., Saurin, W., Quetier, P., Schaffer, M., Muller-Haus, S., Gobel, C., Fuchs, M., Benes, T., Wurzbach, E., Drzonsek, H., Erle, H., Jordan, N., Bangert, S., Wiedemann, R., Kranz, H., Voss, H., Holland, R., Brandt, P., Nyakatura, G., Vezzi, A., D'Angelo, M., Pallavicini, A., Toppo, S., Simionati, B., Conrad, A., Hornischer, K., Kaser, G., Lohrert, T.H., Nordsiek, G., Reichelt, J., Scharf, M., Schom, G., Barges, M., Terol, J., Climent, J., Navarro, P., Collado, C., Perez-Perez, A., Ottenwalder, B., Duchesne, D., Cooke, R., Laudie, M., Berger-Liauro, C., Parnelle, E., Masny, D., de Haan, M., Maize, A.C., Alcaraz, J.P., Cottet, A., Casacuberta, E., Monfort, A., Argiriou, A., Flores, M., Liguori, R., Vitale, D., Mannhaupt, G., Haase, D., Schoof, R., Rudd, S., Zaccaria, P., Mews, H.V., Mayer, K.F., Kaul, S., Town, C.D., Koo, H.L., Talion, L.J., Jenkins, J., Rooney, T., Rizzo, M., Walz, A., Utterbach, T., Fujii, C.Y., Shen, T.P., Crenay, T.H., Haas, B., Maiti, R., Wu, D., Peterson, J., Van Aken, S., Pei, G., Miitscher, J., Sellers, P., Gill, J.E., Feldhlyan, T.V., Preuss, D., Lin, X., Nierman, V.C., Salzberg, S.L., White, O., Vester, J.C., Fraser, C.M., Kaneko, T., Nakamura, Y., Sato, S., Kato, T., Asanuma, E., Sasamoto, S., Kimura, T., Ideawa, K., Kawashima, K., Kishida, Y., Kiyokawa, C., Kohara, M., Matsumoto, M., Matsuno, K., Muraki, A., Nakayama, S., Nakazaki, N., Shimo, S., Takeuchi, C., Wada, T., Watanabe, A., Yamada, M., Yasuda, M. and Takata, S.

CONSTRMEuropean Union Chromosome 3 Arabidopsis Sequencing Consortium; Institute for Genomic Research; Kazusa DNA Research Institute

TITLESquence and analysis of chromosome 3 of the plant Arabidopsis thaliana

JOURNALNature 408 (6814), 820-822 (2000)

PUBMED11130713

REFERENCE2 (residues 1 to 1101)

CONSTRMEBI Genome Project

TITLEDirect Submission

JOURNALSubmitted (19-OCT-2022) National Center for Biotechnology Information, NIH, Bethesda, MD 20894, USA

REFERENCE3 (residues 1 to 1101)

AUTHORSKrishnakumar, V., Cheng, C.-Y., Chan, A.P., Schobel, S., Kim, M., Farauti, E.S., Belyayeva, I., Rosen, B.D., Micklem, G., Miller, J.R., Vaughn, M. and Town, C.D.

Analyze this sequence

Run BLAST

Identify Conserved Domains

Highlight Sequence Features

Find in this Sequence

Show in Genome Data Viewer

Protein clusters for NP_188028.2

Nuclear pore complex protein Nup107-like

Total proteins: 11

Total genes: 0

Conserved in: Embryophyta

Related information

BioProject

Nucleotide

PubMed

Taxonomy

BioSystems

CDD Search Results

Conserved Domains (Concise)

Conserved Domains (Full)

Domain Relatives

Encoding mRNA

Gene

Genome

Protein (UniProtKB)

Protein Clusters

PubMed (RefSeq)

PubMed (Weighted)

Related Structures (List)

LinkOut to external resources

Expression profiles in Arabidopsis thaliana (Protein)

Subcellular localisation in Arabidopsis thaliana (Bio-Analytic Resou)

Transcript/Protein Information (PANTHER Classification Syst)

基本信息记录如下：

蛋白质名称：nuclear pore complex protein

物种：Arabidopsis thaliana (拟南芥)

检索号：NP_188028.2

序列长度：1101 个氨基酸

更新日期：2022 年 10 月 20 日

2.3 点击相关链接获取更全面信息

2.3.1 “FASTA” – 获取完整的氨基酸序列

```
>NP_188028.2 nuclear pore complex protein [Arabidopsis thaliana]
MDMDMDTSPSYFDPEALSVRDQFRYRKRHSTSPHEEMLSSNVSENRLLYDGHNIHSPTNTALLLENIKE
EVDNFHTDHYEGTPTNPISASRRSVGILNDDDEALFRRVESQSLKACKIENDELAESGDTTFALFASLF
DSALQGLMSIPNMLRLEESCRNVSQSIRYGS DIRHRAVEDKLMRQKAQLLLGEAASWSLLWNLYGKGTD
EVPENLILIPSTSHLEACQFVLNDHTAQLCLRIMVWLEELASKSLDERKVQGS HVGYLPNAGVWHHTQ
RYLKKNGSNADTLHHLDFDAPTREHARLLPDDYKQDES VLEDVWTLIRAGRIEEACDLCSAGQSWRAAT
LCPFSGMDMPFSIEALVKNGENRTLQATIEQESGFGNQLRLWKWASYCASEKIAEQDGGKHEVAVFATQCS
NLNRMLPICTDWESACWAMAKSWLDVQVDLELAQSKPGLTERFKSCIDESPEATQNGCQASFGPEDWPLH
VLNQQRDLPALQLKLSGEMVHEAVVRGCKEQHRQIQMNLMLGDISHLLDI IWSWIAPLEDDQSNFRPH
GDPHMIKFGAHMVLVRLRLFTDEINDSFKEKLNNVGD LILHMYAMFLFSKQHEELVGIYASQLARHRCIE
LFVHMMELRMHSSVHVYKIFLSAMEYLSFSPVDDLHGNFEEIVDRVLSRSREIKLAKYDPSIDVAEQHR
QQSLQKAIATQWLCFTPPSTIKDKDVT SKLLRLMHSNILFREFALIAMWRVPATPVGAHTLLSYLAE
PLKQSENPDITLEDYVSENLEQFQDWNEYYS CDAKYRNWLFQLENAEVTELSEENQKAVVAAKETLDS
SLSLLLRQDNPMWTFLEDHVFEEYLFLELHATAMLC LPSGECLRPDATVCAALMSALYSSVSEEVLD
RQLMVNVSISRRSYCIEVLRCLAIKGDGLGPHNANDGGILSAVAAAGFKGSDIYGTYFSFTYDLPPFS
IEIWGCDELTRFQAGVTMDISRDLAWYSSKEGSLETPATYIVRGLCRRCCLPELVLRSMQVSVSLMESGNP
PEDHDELIELVASDETGFLSLFSRQQLQEFMLFEREYRMSQLELQEELSSP
```

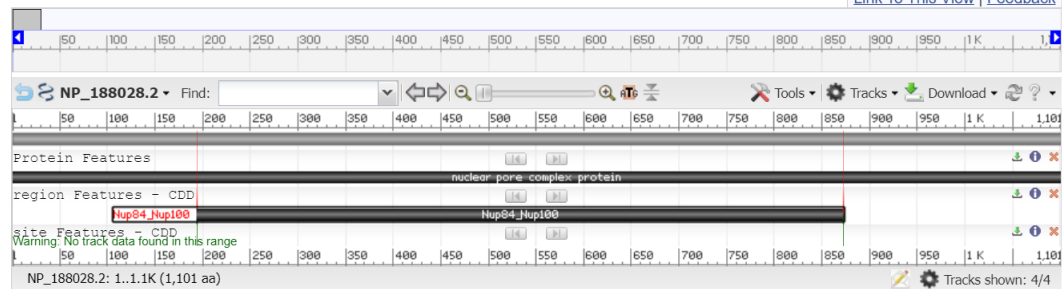
2.3.2 “Graphics” – 查看蛋白质结构域图

nuclear pore complex protein [Arabidopsis thaliana]

NCBI Reference Sequence: NP_188028.2

[GenPept](#) [Identical Proteins](#) [FASTA](#)

[Link To This View](#) | [Feedback](#)



2.3.3 “Identical Proteins” – 查看同源蛋白

nuclear pore complex protein

[GenPept](#) [FASTA](#) [Graphics](#) [BLAST](#)

Name: nuclear pore complex protein RefSeq Selected Product: NP_188028.2, 1101 amino acids Taxonomic Group: eudicots Assembly Accessions: 2 Protein Accessions: 5 CDS Regions: 4 Total Rows: 6				Protein Classes ☒ NP_188028.2 - 2 rows ☒ SwissProt - 1 row ☒ INSDC - 2 rows ☒ Patent - 1 row Update Report		
Source	CDS Region in Nucleotide	Protein	Name	Organism	Strain	Assembly
RefSeq	NC_003074.8 4677993-4685455 (+)	NP_188028.2	nuclear pore complex protein	Arabidopsis thaliana		GCF_000001735.4
RefSeq	NM_112268.3 181-3486 (+)	NP_188028.2	nuclear pore complex protein	Arabidopsis thaliana		
Swiss-Prot	N/A	Q8L748.1	Nuclear pore complex protein NUP107	Arabidopsis thaliana		
INSDC	AY139761.1 180-3485 (+)	AAM98082.1	AT3g14120/MAG2_7	Arabidopsis thaliana		
INSDC	CP002686.1 4677993-4685455 (+)	AEE75472.1	nuclear pore complex protein	Arabidopsis thaliana		GCA_000001735.2
PAT	N/A	ADT59718.1	Sequence 326 from patent US 7847156	Unknown		

2.4 深入分析蛋白质功能

2.4.1 搜索功能信息

在 NCBI 中搜索 "nuclear pore complex arabidopsis", 查找该蛋白质的具体功能和生物学意义, 了解核孔复合体在细胞中的作用。我了解到, 拟南芥核孔复合体是位于植物细胞核膜上的关键蛋白质复合体, 主要负责调控蛋白质、RNA 和其他大分子在细胞核与细胞质之间的双向转运, 其中小分子可自由通过而大分子需要载体蛋白介导, 同时参与基因表达调控、染色质组织和转录因子的核质穿梭过程。在植物特异性功能方面, 核孔复合体深度参与拟南芥的胚胎发育、器官形态建成、开花时间调控以及根系发育等重要生物学过程, 并在植物对光照、温度等环境因子的适应性响应中发挥关键作用, 通过调控抗逆相关基因表达和激素信号转导来帮助植物应对环境胁迫。此外, 该复合体还影响光合作用相关蛋白的细胞定位、次生代谢产物的合成调控以及营养元素代谢, 由约 30 种不同的核孔蛋白组成的精密结构确保了细胞周期的正常进行和 DNA 损伤检查点的信号传导, 其功能异常往往导致胚胎致死、发育缺陷、基因表达紊乱和环境适应能力下降, 因此核孔复合体在维持植物细胞正常生理功能和生存适应性方面具有不可替代的重要意义。

Components of the *Arabidopsis* nuclear pore complex play multiple diverse roles in control of plant growth

[Geraint Parry](#)^{1,*}

[► Author information](#) [► Article notes](#) [► Copyright and License information](#)

PMCID: PMC4203139  **IF: 5.7**  **Q1** PMID: 

该文章指出核孔复合物的成分对多效性生长表型、mRNA 转运和基因表达具有不同的影响, 这在拟南芥核孔蛋白功能的比较研究中得到了体现。该研究集中于拟南芥核孔蛋白突变体的幼苗发育、开花时间、核形态、mRNA 积累和基因表达变化, 揭示了核孔蛋白改变分子和植物生长表型的方式的差异, 表明核孔亚复合物在核转运中发挥着不同的作用, 并揭示了参与核转运的基因表达之间可能存在的反馈关系。其他相关文献情况如下:

1. GBPL3 localizes to the **nuclear pore complex** and **functionally** connects the **nuclear** basket with the nucleoskeleton in plants.
Tang Y, Ho MI, Kang BH, Gu Y.
PLoS Biol. 2022 Oct 21;20(10):e3001831. doi: 10.1371/journal.pbio.3001831^(*).
PMCID: PMC9629626^(*)
The **nuclear** basket (NB) is an essential structure of the **nuclear pore complex** (NPC) and serves as a dynamic and multifunctional platform that participates in various critical **nuclear** processes, including cargo transport, molecular docking, and g ...
2. Nucleoporin PNET1 coordinates mitotic **nuclear pore complex** dynamics for rapid cell division.
Fang Y, Tang Y, Xie P, Hsieh K, Nam H, Jia M, Reyes AV, Liu Y, Xu S, Xu X, Gu Y.
Nat Plants. Author manuscript. 2025 Jan 31;11(2):295-308. doi: 10.1038/s41477-025-01908-y^(*).
PMCID: PMC11850076^(*)
The **nuclear pore complex** (NPC) is a cornerstone of eukaryotic cell **functionality**, orchestrating the nucleocytoplasmic shuttling of macromolecules. ...PNET1 hyperphosphorylation disrupts its interaction with the NPC scaffold, facilitating NPC dismantlin ...
3. Components of the **Arabidopsis nuclear pore complex** play multiple diverse roles in control of plant growth.
Parry G.
J Exp Bot. 2014 Aug 27;65(20):6057-67. doi: 10.1093/jxb/eru346^(*).
PMCID: PMC4203139^(*)
The **nuclear pore complex** (NPC) is a multisubunit protein conglomerate that facilitates movement of RNA and protein between the nucleus and cytoplasm. ...This study reveals differences in the manner by which nucleoporins alter molecular and plant growth phenot ...
4. **Evolutionarily conserved protein motifs drive interactions between the plant nucleoskeleton and nuclear pores.**
Mermert S, Voisin M, Mordier J, Dubos T, Tutois S, Tuffery P, Baroux C, Tamura K, Probst AV, Vanrobays E, Tatout C.
Plant Cell. 2023 Nov 30;35(12):4284-4303. doi: 10.1093/plcell/koad236^(*).
PMCID: PMC10689174^(*)
To understand how the **Arabidopsis** (**Arabidopsis thaliana**) nucleoskeleton physically connects to the **nuclear** periphery in plants, we investigated the **Arabidopsis** nucleoskeleton protein KAKU4 and sought for **functional** regions responsible for its lo ...
5. **Involvement of the nuclear pore complex in morphology of the plant nucleus.**
Tamura K, Hara-Nishimura I.
Nucleus. 2011 May-Jun;2(3):168-72. doi: 10.4161/nucl.2.3.16175^(*).
PMCID: PMC3149876^(*)
Trafficking between the nucleoplasm and the cytoplasm occurs through the **nuclear pore complex** (NPC), which consists of large multiprotein complexes. Over the last several years, major progress has been made in both structural determination of the entire assem ...
6. **The nuclear pore Y-complex functions as a platform for transcriptional regulation of FLOWERING LOCUS C in Arabidopsis.**
Huang P, Zhang X, Cheng Z, Wang X, Miao Y, Huang G, Fu YF, Feng X.
Plant Cell. 2024 Jan 30;36(2):346-366. doi: 10.1093/plcell/koad271^(*).
PMCID: PMC10827314^(*)
The **nuclear pore complex** (NPC) has multiple **functions** beyond the nucleo-cytoplasmic transport of large molecules. ...We show that Y-complex nucleoporins are intimately associated with FLC chromatin through their interactions with histone H2A at ...

2.5 使用其他数据库进行的补充

2.5.1 UniProt (<https://www.uniprot.org/>)

通过 UniProt 这一功能注释最全面的蛋白质数据库，得到了详细的功能描述、结构域信息、翻译后修饰信息。

Functionⁱ

Gene Ontologyⁱ

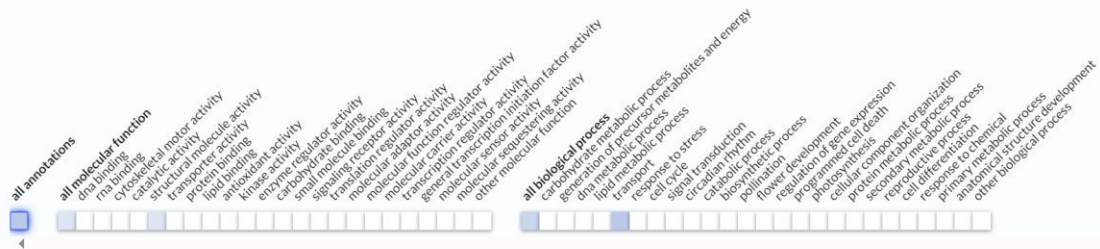
GO annotations

GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Plants



ASPECT	TERM
Molecular Function	structural constituent of nuclear pore ↗ Source:InterPro
Biological Process	mRNA transport ↗ Source:UniProtKB-KW
Biological Process	protein transport ↗ Source:UniProtKB-KW

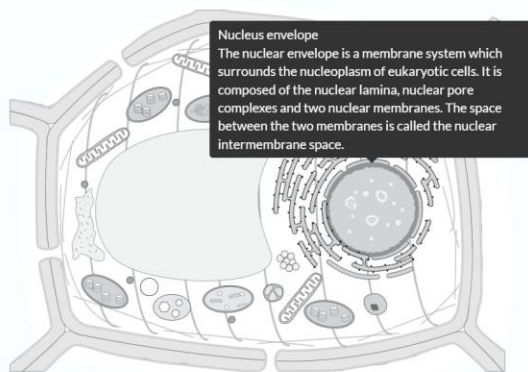
如上图，观察到核孔复合物蛋白的 GO 注释表现主要集中在作为核孔的结构成分、mRNA 转运、蛋白质转运三个方面。而其亚显微结构位置如下图所示。

Subcellular Locationⁱ

UniProt Annotation

GO Annotation

- Nucleus envelope 1 Publication
- Nucleus, nuclear pore complex 1 Publication



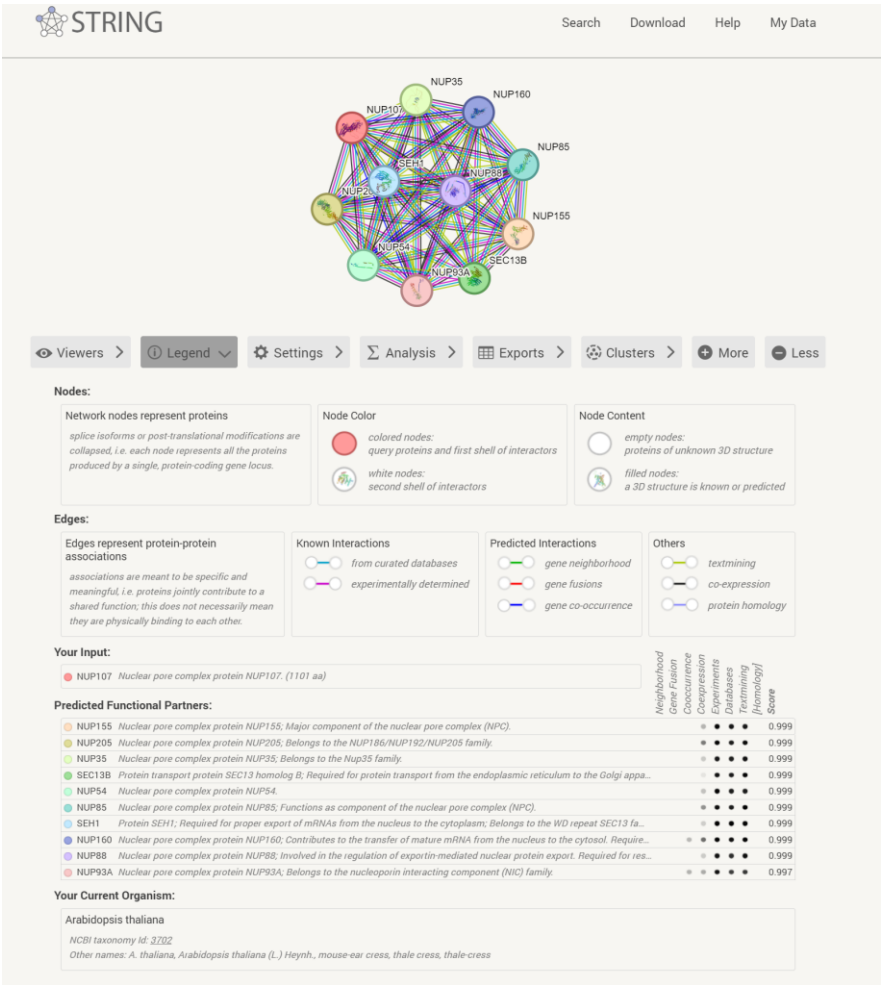
Keywordsⁱ

Cellular component | [#Nuclear pore complex](#)
[#Nucleus](#)

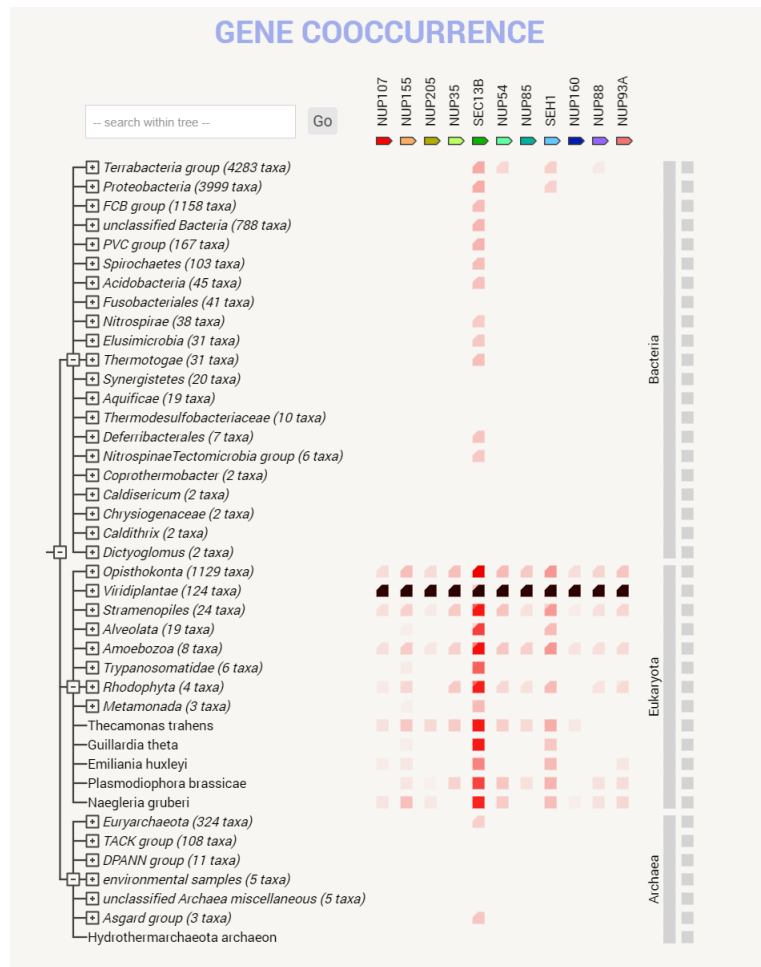


2.5.2 STRING 数据库得到蛋白质相互作用和共表达情况

蛋白质相互作用图如下所示：



基因共表达图如下所示：



三、总结和思考

3.1 核孔复合体在植物细胞中的作用

拟南芥核孔复合体在植物细胞中发挥着多方面的关键作用。作为细胞核与细胞质之间的门户，它精确调控着各类生物大分子的核质转运，包括转录因子的入核、mRNA 的出核以及核糖体亚基的输出等重要过程。同时，核孔复合体还参与基因表达的精细调控，通过影响染色质的空间组织和转录调控因子的定位，间接调节基因的转录活性。在植物特有的生物学过程中，核孔复合体更是参与了光周期调控、激素信号转导和环境胁迫响应等关键生理反应，是植物适应环境变化的重要分子机器。

3.2 该蛋白质的生物学意义

NP_188028.2 编码的核孔复合体蛋白作为核孔复合体的重要组成成分，其生物学意义体现在多个层面。在分子水平上，该蛋白质通过其特有的结构域参与核孔复合体的组装和稳定，确保核质转运的正常进行；在细胞水平上，它影响着细胞周期的进程、DNA 损伤修复和基因

表达调控；在个体水平上，该蛋白质的正常功能直接关系到植物的胚胎发育、器官分化和生理代谢。其功能缺陷往往导致严重的发育异常甚至胚胎致死，凸显了其在植物生命活动中的核心地位。

3.3 进一步研究的可能方向

基于当前的研究基础，该核孔复合体蛋白的后续研究可以从以下几个方向展开：首先，利用结构生物学方法解析该蛋白质的三维结构，深入理解其分子作用机制；其次，通过基因编辑技术构建功能缺失突变体，系统分析其在植物不同发育阶段和环境条件下的具体功能；再次，结合蛋白质组学和转录组学技术，全面解析该蛋白质参与的调控网络和信号通路；最后，比较不同植物物种中同源蛋白的功能差异，探讨核孔复合体蛋白在植物进化中的作用和意义。这些研究不仅有助于深化我们对植物细胞生物学基本规律的认识，也为提高作物的抗逆性和产量提供了潜在的分子靶点。